

Question

A certain mixture **A** is purple-red and contains two common transition metal elements **M**, **N**, with the relative atomic mass of **M** being smaller. The concentrations of these two metals in **A** are close. Concentrated hydrochloric acid is added to **A** and gently heated to obtain a blue solution **B** and a gas **C**. If sodium hydroxide solution is added to **A** with stirring, a brownish-red precipitate **D** and a green solution **E** are obtained. When excess sulfur dioxide is bubbled into **A**, the solution eventually turns into a pink solution **F**. Ammonia water is quickly poured into **F** to obtain a white precipitate **G** and a brownish-yellow solution **H**. **G** slowly turns into a brown precipitate in the air. When **D** and **G** are mixed and dilute sulfuric acid is added, solution **A** is obtained again.

Note: The solution in this problem may contain multiple major ions (ignoring hydrolysis reactions).

Which of the following statements is correct?

- A.** The relative atomic mass of **N** is less than 58.
- B.** The formula weight of a certain major ion contained in **B** is greater than 200.
- C.** **C** is a paramagnetic gas.
- D.** **D** is completely insoluble in dilute hydrochloric acid.
- E.** Solution **E** contains **M** and **N**
- F.** **F** solution contains only **M** or only **N**.
- G.** The central atom of the major ions in **H** has three unpaired electrons.
- H.** None of the above options are correct.

Answer

Correct Answer: **B**

Detailed Explanation

A is a purple-red solution, most likely containing permanganate, indicating that one of the metals is **Mn**;

CHECKPOINT

1 PTS

Purple-red solution most likely contains permanganate, one of the metals is **Mn**

Adding concentrated hydrochloric acid to **A** and heating it will cause the permanganate to oxidize chloride ions to produce Mn^{2+} and chlorine gas **C**, which is light pink; however, the solution is blue, meaning that there are chloride complexes or hydrated ions of other metal ions that are colored, possibly Cu^{2+} , Co^{2+} or Cr^{3+} . Chlorine gas is a diamagnetic molecule, so option C is incorrect.

CHECKPOINT

1 PTS

Permanganate reduces chloride ions to produce Mn^{2+} and gas **C** Cl_2

CHECKPOINT

1 PTS

The presence of chloride complexes or hydrated ions of other metal ions gives a blue color, possibly Cu^{2+} , Co^{2+} or Cr^{3+}

Adding sodium hydroxide solution to **A** will convert the permanganate into manganate and manganese dioxide, with manganate being a green solution **E** and manganese dioxide being a brown precipitate **D**.

CHECKPOINT

1 PTS

Permanganate in alkaline conditions will be converted to green solution **E** MnO_4^{2-} and brown precipitate **D** MnO_2

Passing sulfur dioxide into **A** to generate **F** will reduce the permanganate to Mn^{2+} . Since solution **F** is pink, it indicates that the reduced state of the other metal ion is also pink or very lightly colored; Co^{2+} is exactly pink, so it is guessed that the other metal is **Co**.

CHECKPOINT

1 PTS

The reduced state of the other metal ion is also pink or very lightly colored

CHECKPOINT

1 PTS

Guessing that the other metal is **Co**

Adding sodium hydroxide solution to **A** converts the other metal ion into a brown precipitate **D**. If the metal is Cu^{2+} , its hydroxide should be blue; if the metal is Cr^{3+} , its hydroxide is gray-green; if the metal is **Co**, the generated cobalt(III) hydroxide is exactly brown, confirming the above guess; therefore, it is determined that **M** is **Mn** and **N** is **Co**. The molecular weight of **Co** is greater than 58, so option A is incorrect.

CHECKPOINT

1 PTS

$\text{Co}(\text{OH})_3$ is brown

CHECKPOINT

1 PTS

M is **Mn**, **N** is **Co**

At this time, the colored chloride complex in **B** is CoCl_4^{2-} , with a formula weight of 200.9, which is greater than 200, so option B is correct.

CHECKPOINT

1 PTS

The colored chloride complex in **B** is CoCl_4^{2-}

The brown precipitate **D** contains manganese dioxide and cobalt(III) hydroxide. Cobalt(III) hydroxide is soluble in dilute hydrochloric acid, so option D is incorrect.

CHECKPOINT

1 PTS

The brown precipitate **D** contains MnO_2 and $\text{Co}(\text{OH})_3$

Solution **E** only contains manganate, and Co^{2+} is converted into cobalt(III) hydroxide, so option E is incorrect.

CHECKPOINT

1 PTS

Solution **E** only contains MnO_4^{2-}

F contains Mn^{2+} and Co^{2+} , both of which are pink, so option F is incorrect.

CHECKPOINT

1 PTS

F contains Mn^{2+} and Co^{2+}

Quickly pouring ammonia into **F** yields a white precipitate **G** and a brownish-yellow solution **H**. The white precipitate **G** is $\text{Mn}(\text{OH})_2$, so **H** is the ammine complex of Co^{2+} , $\text{Co}(\text{NH}_3)_6^{2+}$, which has a d^7 configuration. Whether it is low-spin or high-spin, there must be Jahn-Teller distortion, so option G is incorrect.

CHECKPOINT

1 PTS

White precipitate **G** is $\text{Mn}(\text{OH})_2$

CHECKPOINT

1 PTS

H is $\text{Co}(\text{NH}_3)_6^{2+}$, which has a d^7 configuration, there must be Jahn-Teller distortion

In summary, option B is correct.